Supplementary Material

Measured Effects and Limits of Inference-Time Geometry-Adapter Scaling for Multi-view Texture Generation

1 S1. FRESH_CONFIRM_B generic schedule comparisons

The generic schedule extension was executed on the same 150-object B cohort after the original analysis had been opened. It passed its 1,200/1,200-row integrity gate, but is a same-cohort post-lock sensitivity and not an independent confirmation. Each entry is LLH minus the comparator. Positive FG-PSNR and negative FG-LPIPS favor LLH. Holm correction is separate across the eight schedule comparisons for each metric.

Table 1: Endpoint-specific generic schedule comparisons. Intervals are paired object-bootstrap 95% CIs.

Schedule	FG-PSNR Δ [95% CI]; Holm p	FG-LPIPS Δ [95% CI]; Holm p
Generic linear	-0.0373 [-0.0829, 0.0086]; 0.1150	+0.0011 [0.0004, 0.0019]; 0.0024
Linear, requested-mean matched	+0.0967 [0.0594, 0.1357]; ≤0.0008	-0.0009 [-0.0013, -0.0004]; 0.0008
Cosine bump	+0.3367 [0.2216, 0.4576]; ≤0.0008	-0.0047 [-0.0059, -0.0034]; ≤0.0008
Cosine bump, requested-mean matched	+0.4629 [0.3526, 0.5780]; ≤0.0008	-0.0059 [-0.0071, -0.0046]; ≤0.0008
Trapezoid	+0.1875 [0.0697, 0.3099]; 0.0036	-0.0035 [-0.0047, -0.0023]; ≤0.0008
Trapezoid, requested-mean matched	+0.4404 [0.3334, 0.5521]; ≤0.0008	-0.0057 [-0.0070, -0.0045]; ≤0.0008
Gaussian peak	+0.4092 [0.2934, 0.5302]; ≤0.0008	-0.0052 [-0.0065, -0.0040]; ≤0.0008
Gaussian peak, requested-mean matched	+0.4711 [0.3584, 0.5892]; ≤0.0008	-0.0058 [-0.0071, -0.0045]; ≤0.0008

Requested-mean matching does not equalize measured correction norms. The nine-method total-dose central-90% intersection is empty. Pairwise dose-conditioned models added after outcome exposure were underpowered or had insufficient overlap; they do not identify a dose-adjusted schedule-shape advantage. The complete condition-level cap and dose accounting is provided in `evidence/audits/FINAL_CAP_BUDGET_CAUSAL_ACCOUNTING.md` and `evidence/audits/FINAL_GENERIC_SCHEDULE_COMPARISON.md`.

2 S2. Multi-seed realization sensitivity

E1 averages each object's paired effect over fixed generation seeds 42, 43, and 44 before object-level inference. Forty-eight objects were reused from B. The rows below show the LLH–LFM-exact comparison.

Table 2: E1 effect across three fixed generation seeds.

Quantity	FG-PSNR (dB)	FG-LPIPS
Mean object-averaged effect [95% CI]	+0.329 [0.179, 0.497]	-0.00391 [-0.00549, -0.00245]
Favorable-object fraction	75.0%	79.2%
Seed 42 mean	+0.284	-0.00451
Seed 43 mean	+0.605	-0.00516
Seed 44 mean	+0.100	-0.00205
Effect / median within-object seed range	0.58	0.68

The seed-range ratios are descriptive scale comparisons only. The confidence intervals quantify object resampling conditional on these three generation seeds, not a random-seed population interval. Seed 42 reuses B outputs, and this subset was not a new independent cohort.

3 S3. E2 static 2-by-2 profile details

The locked static profiles were F00=(1.25, 1.25, 0.50), F01=(1.25, 1.25, 0.80), F10=(1.675, 1.675, 0.50), and F11=(1.675, 1.675, 0.80), in deep/middle/shallow order. Each profile is constant for all 50 steps. The deep and middle scales are a single combined factor. The object-level factorial effects average paired contrasts over seeds 42–44 before inference.

The primary interaction is not a temporal interaction. The formal design and row-accounting audit verify 576 logical rows (480 newly generated and 96 verified seed-42 reuses), 216,000 residual-trace entries, and 1,152

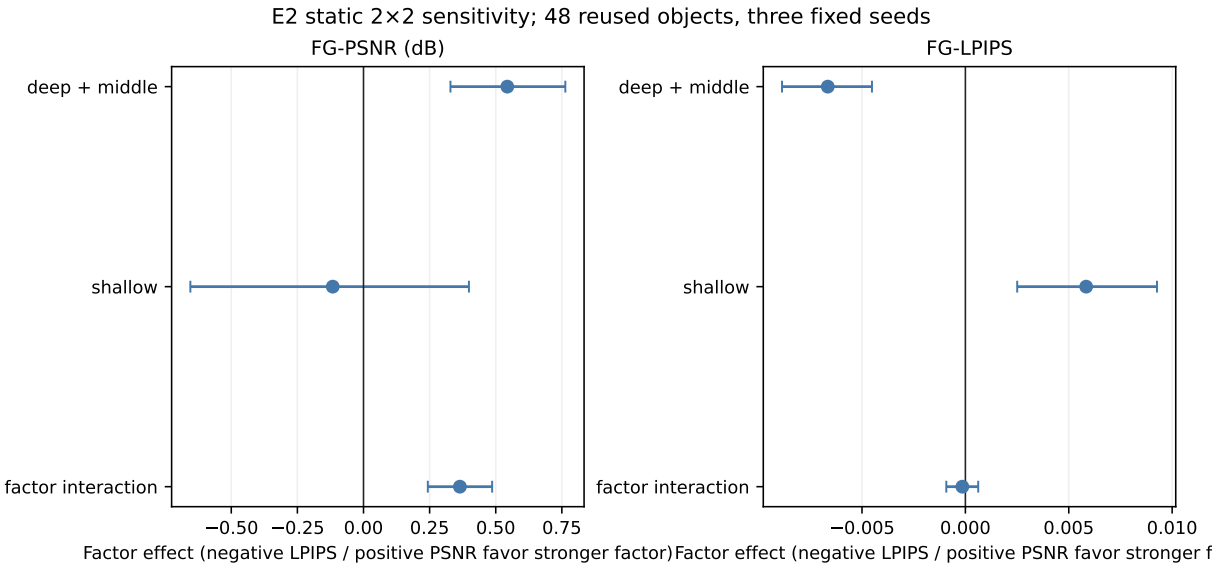

Figure 1: Object-bootstrap estimates for the three E2 factorial effects. The deep+middle-by-shallow interaction is endpoint-specific.

prediction/trace file hashes. The seed-42 F01 label requested 1.25 for shallow but the native cap applied 0.80; its predictions and effective trace match the 0.80 request. This request-field discrepancy is preserved in the alias erratum.

4 S4. Bounded layer-by-window map and dose limits

A3b covers 15 layer-by-window interventions relative to the all-low reference. The cluster-Wald and GEE tests detect response heterogeneity (FG-LPIPS interaction RMSE 0.000342; FG-PSNR 0.0205 dB). The dose calibration target was missed: deep and middle were approximately 20% below target, while shallow was 98.98% above target. The adjusted three-layer dose model retains 0/2,250 rows in common support. Therefore the response-surface test is not interpreted as a dose-independent mechanism. A2 is discovery evidence and is not used as an independent confirmatory replication.

5 S5. GLB-native evaluation and visual audit

The stored GLBs were rendered with their embedded glTF base-color sampler and frozen camera protocol. Source GLB hashes (160) and rendered image hashes (1,760) passed. Each object mean pools 11 unseen views. No texture was rebaked. The endpoint is unlit base-color and does not establish full-PBR appearance. Four objects without UVs were excluded before baking, so strict N=24 3D coverage is unsupported. The old seam analyzer is not treated as valid evidence for the exported mapping; no seam claim is made.

The 24-object 2D visual cohort was fixed using ground-truth-only strata before method outcomes. Each panel compares the same six target views for ground truth and seven generated conditions, including no adapter and fixed-scale controls. The panels are a completeness-oriented visual record; they are not a blinded human rating. Manual inspection recorded material/color mismatch in 18/24 objects and fine-detail loss in 21/24. Individual source-object attribution appears in S10.

Metric-ranked strongest-win, median-effect, and strongest-loss panels are available in `manuscript/figures/`; they are selected illustrations from the 24-object visual cohort, not independent evidence. Their rank rule and object IDs are in the copied visual-evidence report.

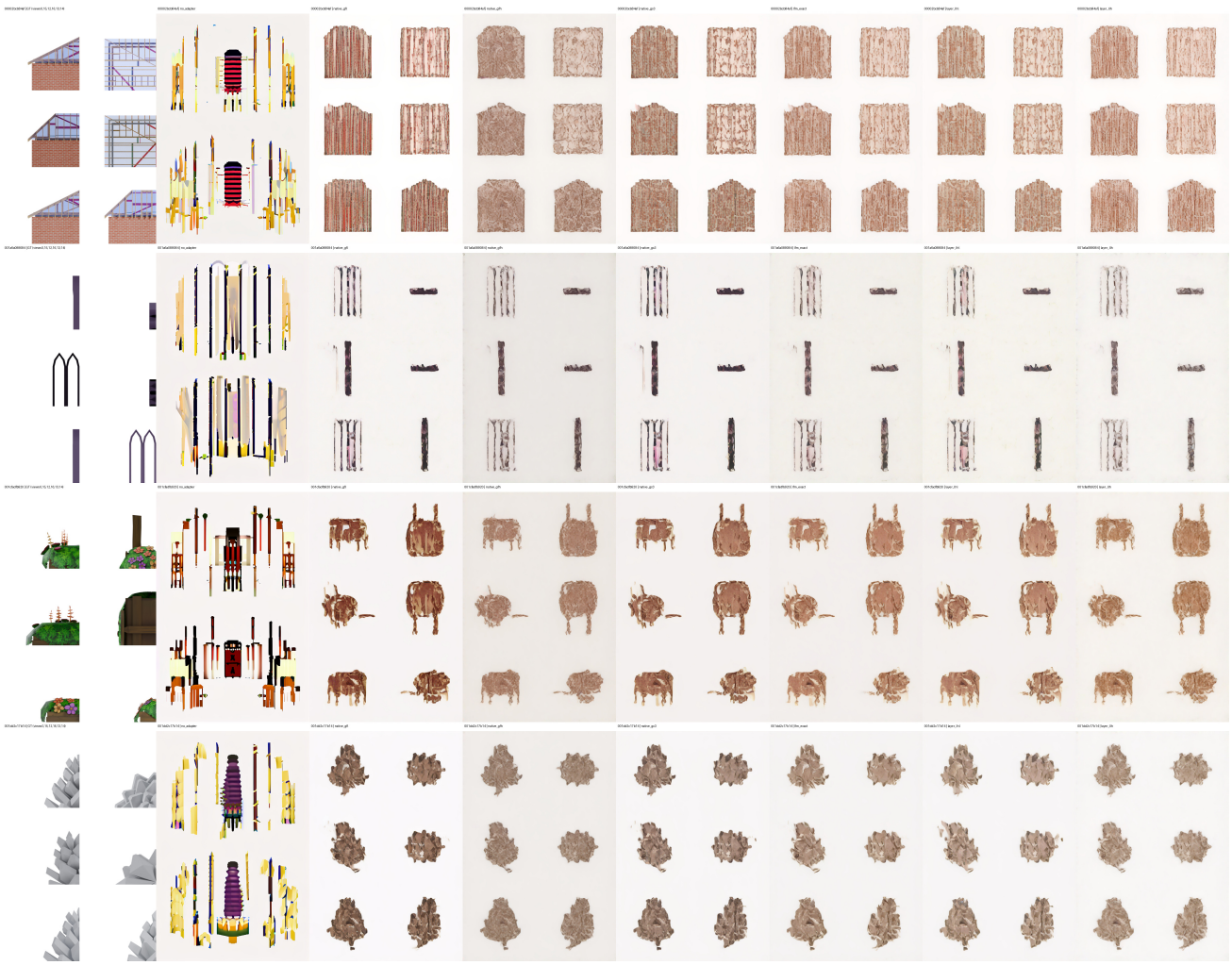

Figure 2: Preselected GT-stratified visual cohort, objects 1–4 of 24. Each row contains the fixed target views and same-draw outputs for the unmodified pipeline, native fixed-low and fixed-high baselines, GC3, LFM-exact, LHL, and LLH. Descriptive only; columns do not imply a ranking and do not evaluate the baked GLB endpoint.

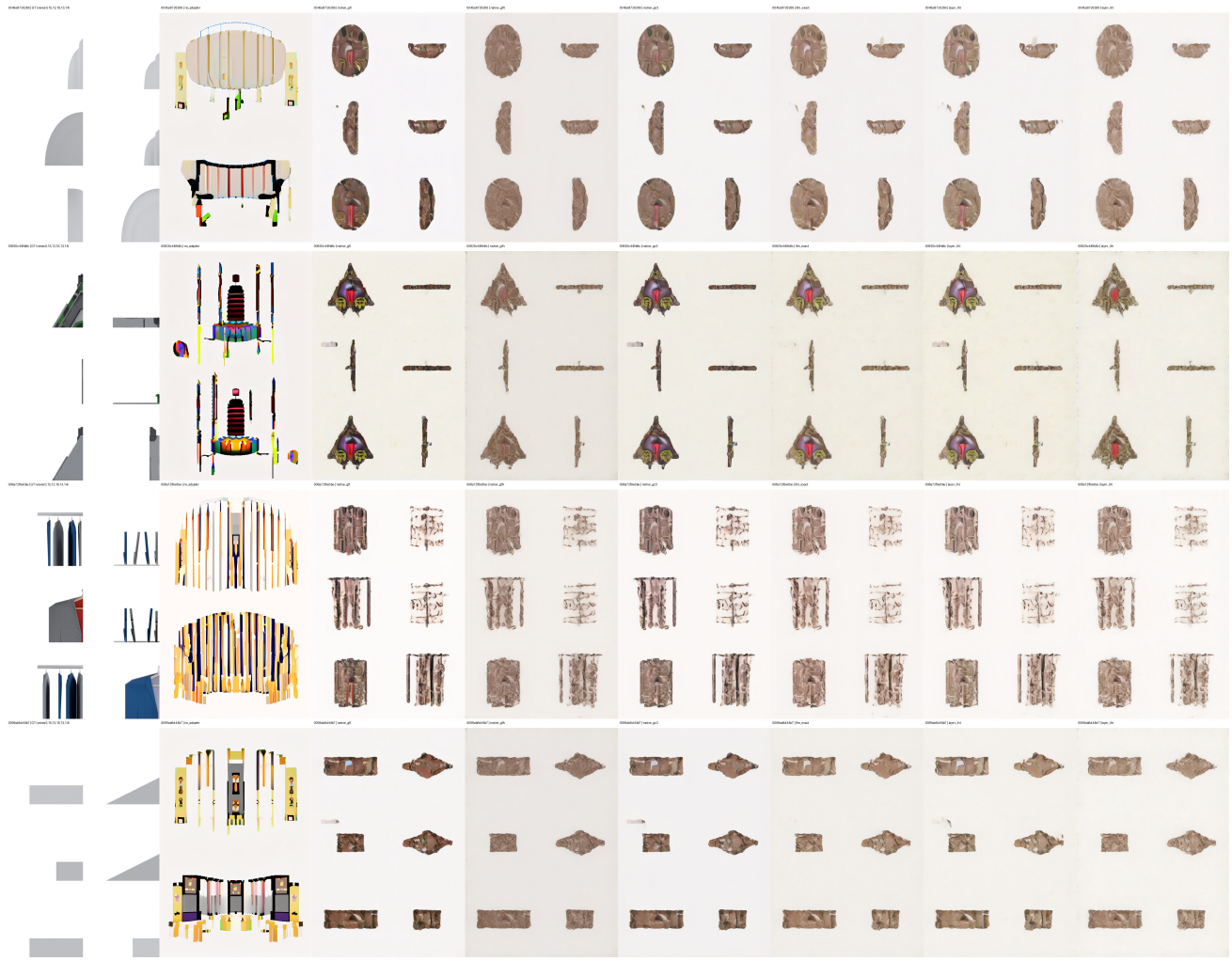

Figure 3: Preselected GT-stratified visual cohort, objects 5–8 of 24. Conditions and fixed target views match Fig. S1.

Figure 4: Preselected GT-stratified visual cohort, objects 9–12 of 24. Conditions and fixed target views match Fig. S1.

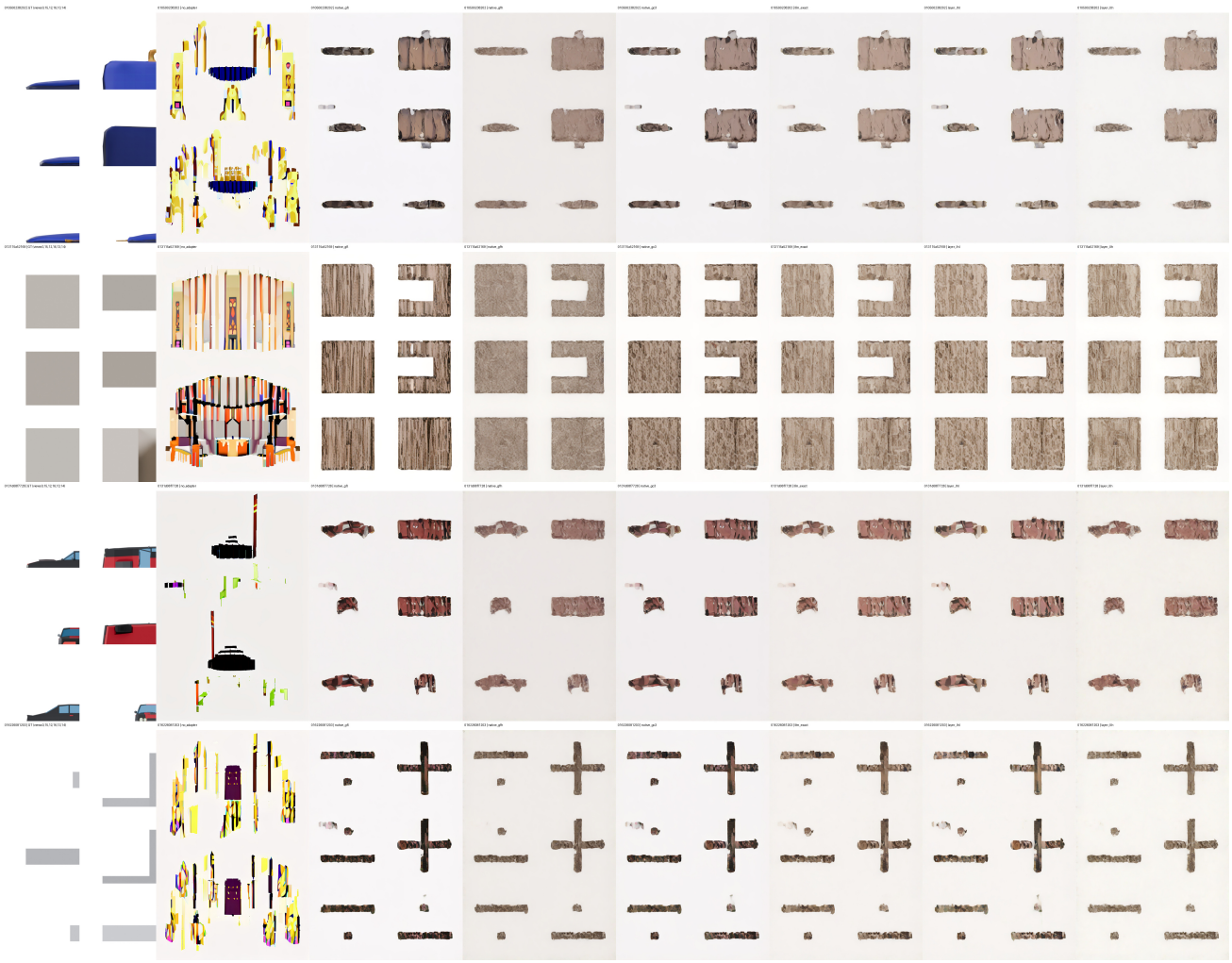

Figure 5: Preselected GT-stratified visual cohort, objects 13–16 of 24. Conditions and fixed target views match Fig. S1.

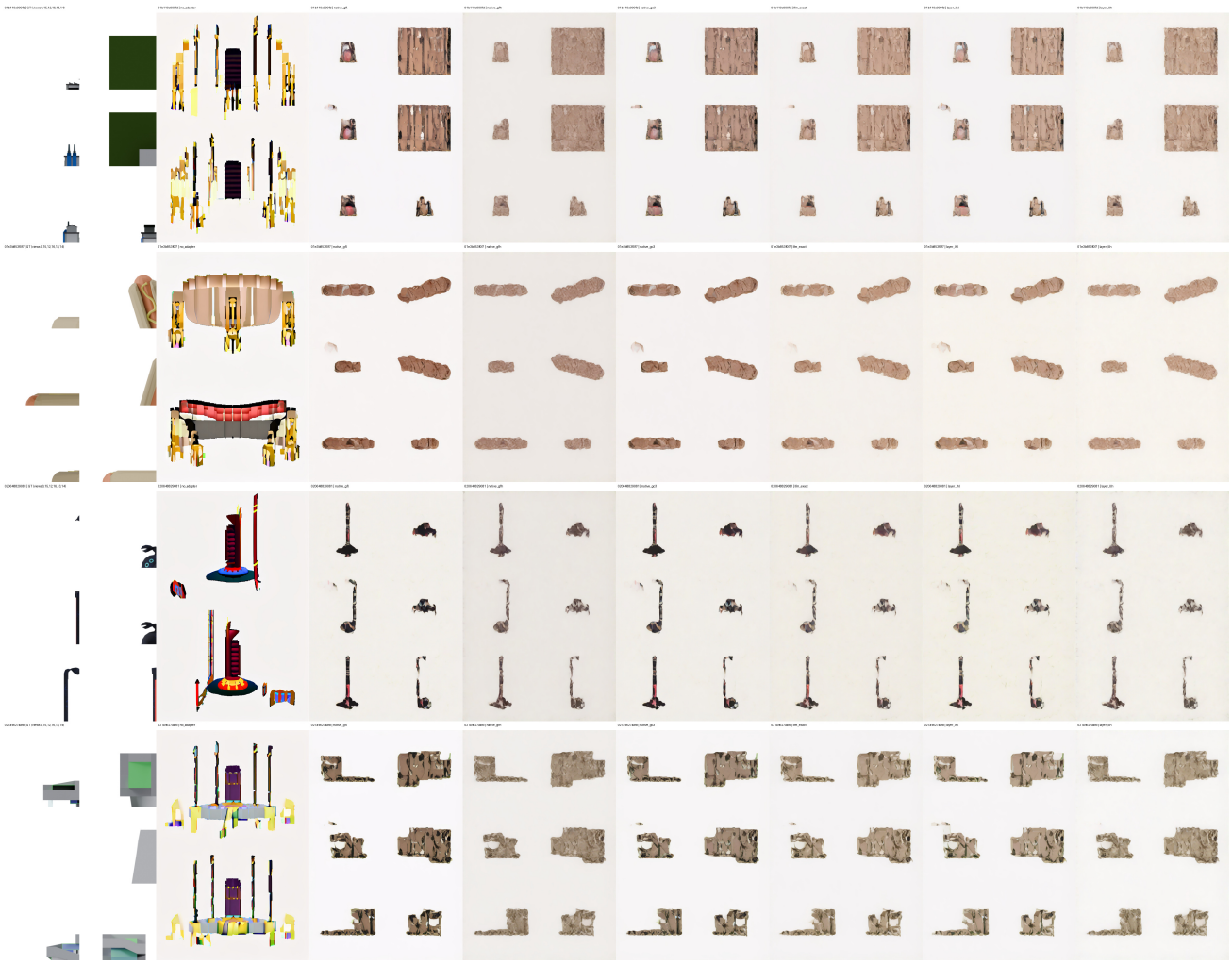

Figure 6: Preselected GT-stratified visual cohort, objects 17–20 of 24. Conditions and fixed target views match Fig. S1.

Figure 7: Preselected GT-stratified visual cohort, objects 21–24 of 24. Conditions and fixed target views match Fig. S1.

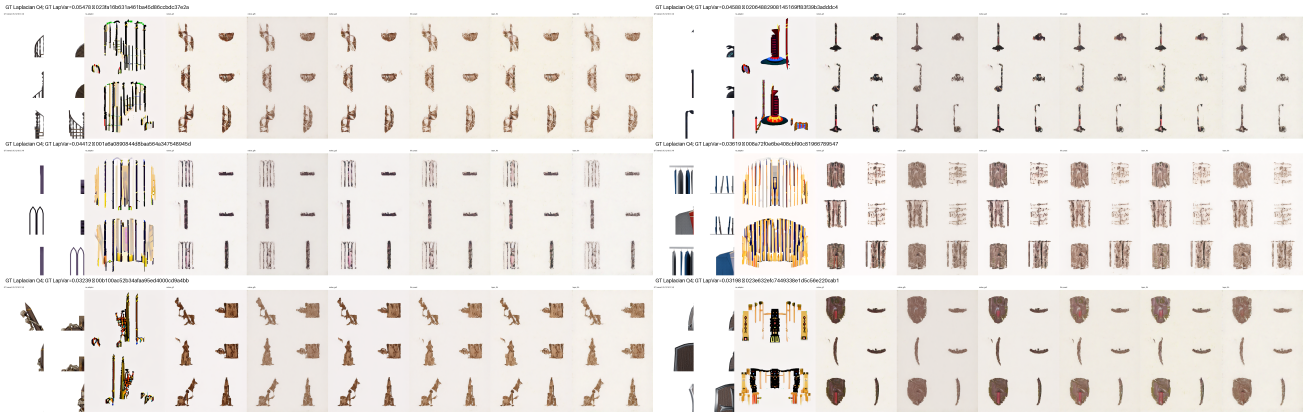

Figure 8: Six highest ground-truth Laplacian-variance objects within the same fixed 24-object cohort, shown across the recorded conditions. This outcome-independent subset is descriptive and does not estimate a population failure rate.

6 S6. Retrospective C3 paired re-evaluations

All estimates in S6 are retrospective object-paired supplements. The strict-276 GC3–GFL and GC3–GFH pairs were not registered primary Core-7 contrasts; legacy GFL/GC3 input-tensor hashes and common-runner identity are incomplete. FRESH_CONFIRM_B is a disjoint 150-object cohort, but it was already unblinded and the native-GC3–native-GFL B3 pair was not registered. The B pair therefore has an independent object set but is post hoc, not an independent prospective or registered confirmation. Every interval below is an unadjusted, nominal percentile object-bootstrap 95% CI; favorable-object direction is high for PSNR, SSIM, and Edge-SSIM, and low for LPIPS. Exact object-level deltas and analysis scripts are included in `data/derived/` and `scripts/`.

On strict-276, all seven GC3–GFL endpoint directions favor GC3, whereas only full-image PSNR and SSIM favor GC3 over fixed-high; foreground and edge metrics favor fixed-high. On the disjoint B objects, all seven mean directions favor GC3 over GFL, but this post-hoc analysis cannot erase the earlier exposure or be treated as a fresh registered replication. The historic pooled +0.96 dB is not carried forward as an independently confirmed effect.

7 S7. Cross-interface results

The MV-Adapter experiment evaluates 98 objects from 99 planned across a bounded one-window additive-residual intervention. No global interaction survives the exploratory four-metric correction; isolated late shallow LPIPS changes are small and do not establish a general null or a transfer law.

MVDiffusion is a separate 75-object result on the unchanged SD2-depth compatibility deployment base. Its schedule interpolates serial correspondence-aware CPBlock outputs, not an independent geometry-residual branch. For layer-LLH versus global fixed-low, paired means trade structural/color endpoints against GT-relative texture error: PSNR is -0.454 dB, FG-SSIM -0.0194, Edge-SSIM -0.0059, while CIEDE2000 is -1.373; FG-LPIPS is not separated. This is a boundary for that interface and intervention, not a matched causal architecture comparison. Absolute scores are never pooled across these cohorts.

8 S8. Human preference and scope

The original 01549 3AFC study is retained as complementary perceptual-preference evidence for its original conditions only. It included 24 participants and 30 objects; each criterion had 720 choices. As reported in the submitted study, the uniform-scale 1.25 condition received 46.1% of texture-naturalness choices, uniform-scale 2.50 received 46.0% of shape-consistency choices, and C3/TCAS received 58.1% of overall-quality choices (reported 95% participant-by-object cluster-bootstrap CI [50.1%, 65.8%]). This pattern supports a preferred balance among those three conditions, not universal superiority or transfer to LLH. The raw vote-level file was unavailable for an independent reanalysis; the reported counts and interval are therefore identified as results from the submitted 01549 study.

We analyzed the prespecified 40-slot human-preference study of LLH versus GFL, GFH, GC3, and generic linear. The appearance question asked which output better matched the reference object's visible appearance, including color and material; the texture question asked about texture naturalness. Forty slots were represented. We retained 37 field-valid participants (two failed the recorded comprehension check and one response was incomplete), covering 24 objects and eight comparison-by-question endpoints. For each endpoint, judgments were scored as LLH choice = 1, comparator choice = 0, tie = 0.5; object means were equally weighted, and uncertainty was estimated with a two-way participant/object cluster bootstrap (10,000 resamples, fixed seed 20261005). The eight pre-specified endpoints formed one Holm family. Pointwise 95% intervals are shown; the number of judgments is 222 per endpoint. This is confirmatory for the planned endpoint family, with the execution deviations below reported in full.

Table 5: Prespecified confirmatory human-preference endpoints with documented execution deviations. Values above 0.5 favor LLH descriptively; no endpoint survives Holm correction.

Comparator	Question	LLH preference [95% CI]	Raw p	Holm p
GFL	Appearance match	0.591 [0.459, 0.718]	0.174	0.790
GFL	Texture naturalness	0.647 [0.528, 0.755]	0.012	0.098
GFH	Appearance match	0.567 [0.447, 0.689]	0.272	0.815
GFH	Texture naturalness	0.603 [0.488, 0.714]	0.074	0.442
GC3	Appearance match	0.587 [0.464, 0.709]	0.158	0.790
GC3	Texture naturalness	0.632 [0.521, 0.741]	0.018	0.128
Generic linear	Appearance match	0.540 [0.412, 0.667]	0.536	1.000
Generic linear	Texture naturalness	0.539 [0.420, 0.653]	0.511	1.000

Although all eight point estimates exceed 0.5, none provides multiplicity-adjusted evidence that participants preferred LLH. In particular, the generic-linear intervals cross 0.5; absent an equivalence margin and with the intervals spanning both directions, these results do not establish equivalence or non-inferiority. The four comparisons and eight-endpoint analysis were specified before collection, so we report the endpoint analysis as confirmatory with documented execution deviations. None of the 40 uploaded task schedules matches a schedule in the frozen assignment package; left/right orientation is fixed within all 96 object-by-comparison cells across participants, although LLH is left for 12 and right for 12 objects in each comparator; and the export lacks a randomization seed, question order, raw comprehension answers, and a stimulus-alias-to-frozen-image hash crosswalk. The response export also lacks the coordinator's completed/withdrawn closeout log and ethics/consent documentation. These deviations limit verification of randomization, stimulus identity, stopping, and study governance; they do not justify discarding the prespecified endpoints or selectively retaining results. The study measures relative judgments of fixed 2D outputs and does not establish absolute reference fidelity, unseen-view quality, or 3D-bake quality.

The aggregate endpoint table, forest plot, analysis provenance, source-commit record, and executable analysis script are included under `evidence/human_study/`. The four anonymous participant-level CSVs were previously uploaded at the source branch and commit recorded in the provenance file; this package does not copy them again. Their continued public availability and the missing consent/ethics documentation remain separate data-governance items.

9 S9. Provenance and supersession

The first authority for claim wording is `evidence/audits/next_stage_20261006/NARRATIVE_DECISION_MAP.md` together with the hash-validated claim-authority ledger. The GLB-native report supersedes the older CPU-renderer or seam summaries for stored exported GLBs. Fresh B is a disjoint object cohort but its lock chronology is retrospective. E1/E2 are fixed-subset sensitivity analyses. The strict-276 C3 contrast is retrospective. Full generated image and residual-trace outputs are omitted from this GitHub evidence copy because of size and redistribution constraints; their source hashes and original local paths remain indexed in the original audit manifests. Do not treat omitted payload hashes as a substitute for access to the payloads when a clean-clone reproduction is required.

10 S10. Per-object visual asset attribution

The 24-object visual cohort uses Objaverse 1.0 assets. The table maps each panel to its source UID and creator, and links the Sketchfab model page and Creative Commons license. Current API metadata corroborates 22/24 license records. For panel 02 the current API license field is empty; for panel 05 the model endpoint returns 404. Both retain the Objaverse snapshot `by` slug and are marked with †. The stored description for panel 02 also refers to personal-project use, so its redistribution terms need author-level confirmation before the final public release. The ShareAlike asset in panel 06 is separately identified; any redistribution of an adaptation must honor its terms. No source meshes are included in this package.

Table 6: Source and license attribution for the fixed visual panel.

Panel	Source UID / model page	Creator	License	API record
00	Sketchfab model page	mstam	CC BY 4.0	live API
01	Sketchfab model page	UltraMelon	CC BY 4.0	live API
02	Sketchfab model page	johnguataquira	CC BY 4.0†	snapshot only†
03	Sketchfab model page	IronEqual	CC BY 4.0	live API
04	Sketchfab model page	AVRORA Group	CC BY 4.0	live API
05	Sketchfab model page	Flomezsupercool3 / coolpurpleguy	CC BY 4.0†	snapshot only†
06	Sketchfab model page	1-3D.com	CC BY-SA 4.0	live API
07	Sketchfab model page	127469	CC BY 4.0	live API
08	Sketchfab model page	MARTINICE GROUP	CC BY 4.0	live API
09	Sketchfab model page	DocStephen	CC BY 4.0	live API
10	Sketchfab model page	jorgebaby	CC BY 4.0	live API
11	Sketchfab model page	tsutomu.saito	CC BY 4.0	live API
12	Sketchfab model page	astridgarod	CC BY 4.0	live API
13	Sketchfab model page	GmedranoTIC	CC BY 4.0	live API
14	Sketchfab model page	Wordofcurse	CC BY 4.0	live API
15	Sketchfab model page	szaboo	CC BY 4.0	live API
16	Sketchfab model page	SANYABEAST	CC BY 4.0	live API
17	Sketchfab model page	IronEqual	CC BY 4.0	live API
18	Sketchfab model page	samuelPaq	CC BY 4.0	live API
19	Sketchfab model page	marvin	CC BY 4.0	live API
20	Sketchfab model page	ahmadvicky_zent	CC BY 4.0	live API
21	Sketchfab model page	wolfgar74	CC BY 4.0	live API
22	Sketchfab model page	Dinesh Naidu	CC BY 4.0	live API
23	Sketchfab model page	nbadilan	CC BY 4.0	live API

Table 3: Strict-276 retrospective GC3 paired differences. Delta is GC3 minus the comparator.

Endpoint	GC3 − GFL mean [95% CI]	Favorable	GC3 − GFH mean [95% CI]	Favorable
FG-PSNR (dB)	+1.208 [1.117, 1.295]	259/276	-1.214 [-1.314, -1.115]	25/276
FG-SSIM	+0.03086 [0.02832, 0.03335]	261/276	-0.04878 [-0.05154, -0.04590]	7/276
FG-LPIPS	-0.01003 [-0.01093, -0.00912]	248/276	+0.01396 [0.01298, 0.01491]	13/276
Full-PSNR (dB)	+0.7389 [0.6476, 0.8275]	226/276	+0.7252 [0.6471, 0.8065]	240/276
Full-SSIM	+0.007184 [0.006571, 0.007781]	253/276	+0.001229 [0.000570, 0.001886]	187/276
Full-LPIPS	-0.007819 [-0.008780, -0.006851]	228/276	+0.006176 [0.004968, 0.007341]	76/276
Edge-SSIM	+0.007760 [0.006773, 0.008746]	233/276	-0.02601 [-0.02804, -0.02399]	18/276

Table 4: FRESH_CONFIRM_B post-hoc native-GC3 minus native-GFL differences. Holm p values are an exploratory sensitivity across seven endpoints and do not make the contrast registered.

Endpoint	Mean difference [nominal 95% CI]	Favorable objects	Post-hoc Holm p
FG-PSNR (dB)	+0.502 [0.342, 0.663]	100/150	0.0007
FG-SSIM	+0.02380 [0.01911, 0.02901]	123/150	0.0007
FG-LPIPS	-0.00513 [-0.00655, -0.00369]	107/150	0.0007
Full-PSNR (dB)	+0.225 [0.095, 0.358]	82/150	0.0007
Full-SSIM	+0.005934 [0.004797, 0.007107]	130/150	0.0007
Full-LPIPS	-0.004548 [-0.006102, -0.003009]	109/150	0.0007
Edge-SSIM	+0.005034 [0.003348, 0.006753]	105/150	0.0007